

RAID Configuration and Data Redundancy

Task 1 – RAID Comparison

Question:

Create a comparison table listing RAID 0, RAID 1, RAID 5, and RAID 10, including minimum number of disks, data redundancy, speed, and a real-world use case.

Answer:

RAID 0 uses a minimum of 2 disks and provides no redundancy. It uses striping, so it offers high read/write performance, but a single disk failure can cause loss of the entire array. A suitable use case is temporary high-speed storage or video editing where backups are available.

RAID 1 uses a minimum of 2 disks and provides high redundancy through mirroring. The same data is written to both disks. Read performance can be good, while write performance is generally similar to a single disk. It is suitable for important data such as application or file-server storage where protection from a single disk failure is important.

RAID 5 uses a minimum of 3 disks and provides redundancy through distributed parity. It offers good read performance and efficient usable capacity, while write performance is lower than RAID 0 because parity must be calculated. It is commonly suitable for file servers and storage systems.

RAID 10 uses a minimum of 4 disks and combines mirroring and striping. It provides strong redundancy and high read/write performance, but only about half of the raw capacity is usable. It is suitable for workloads such as databases, virtualization, and high-performance application servers.

Comparison:

RAID 0 | Minimum disks: 2 | Redundancy: None | Speed: Very high | Use case: High-speed temporary/video storage

RAID 1 | Minimum disks: 2 | Redundancy: High, mirroring | Speed: Good | Use case: Important file/application storage

RAID 5 | Minimum disks: 3 | Redundancy: Single-disk failure protection | Speed: Good reads, moderate writes | Use case: File servers

RAID 10 | Minimum disks: 4 | Redundancy: High, mirroring + striping | Speed: Very high | Use case: Databases and virtualization

Task 2 – RAID 1 Data Redundancy

Question:

Explain with your own example how RAID 1 provides data redundancy using a scenario like storing WhatsApp chat backups on two disks.

Answer:

Suppose a computer stores WhatsApp chat backups on a RAID 1 array containing Disk 1 and Disk 2. When a backup file is saved, RAID 1 writes the same data to both disks. Therefore, both disks contain an identical copy of the backup.

If Disk 1 fails, the RAID array can continue operating from Disk 2 because Disk 2 contains the same data. The backup remains accessible even though one disk has failed. The failed disk can then be replaced and the RAID 1 array can rebuild the mirror.

This demonstrates that RAID 1 improves data availability by keeping a duplicate copy of the data on another disk. However, RAID 1 is not a substitute for a separate backup because accidental deletion or corruption can also be mirrored.

Task 3 – Simulate RAID 1 with VirtualBox and Ubuntu

Question:

Create two virtual hard disks, use mdadm to configure them in RAID 1, and check /proc/mdstat.

Answer:

Procedure:

1. Create an Ubuntu virtual machine in VirtualBox.
2. Add two additional virtual hard disks of the same size to the VM.
3. Start Ubuntu and identify the two disks using:

```
lsblk
```

4. Install mdadm:

```
sudo apt update
```

```
sudo apt install mdadm
```

5. Create the RAID 1 array using the two disks. For example, if the disks are /dev/sdb and /dev/sdc:

```
sudo mdadm --create --verbose /dev/md0 --level=1 --raid-devices=2 /dev/sdb /dev/sdc
```

6. Check the RAID status:

```
cat /proc/mdstat
```

7. Verify the array details:

```
sudo mdadm --detail /dev/md0
```

Expected result:

The /proc/mdstat output should show an active md0 RAID 1 array with two devices and indicate that the array is synchronizing or has completed synchronization.

Screenshot evidence:

The assignment asks for a screenshot of the actual `cat /proc/mdstat` output. This document provides the correct procedure and expected result; an actual screenshot should be taken from the Ubuntu VM if evidence is required.

Task 4 – RAID 1 Disk Failure Test

Question:

Delete a file from the RAID 1 array, power off one virtual disk to simulate failure, and check whether the file is still accessible.

Answer:

1. Create or copy a test file into the mounted RAID 1 filesystem.
2. Confirm that the file can be opened normally.
3. Simulate a failure of one of the RAID disks by making one virtual disk unavailable.
4. Check the RAID status using:

```
cat /proc/mdstat
```

5. Access the mounted RAID filesystem and verify that the test file is still available.
6. The RAID 1 array should remain accessible in a degraded state because the remaining disk contains a mirror of the data.

Observation:

When one disk fails, RAID 1 continues to provide access to the mirrored data on the surviving disk. The array will normally report a degraded state until the failed disk is replaced and the mirror is rebuilt.

Note:

Deleting a file is a data operation and is normally mirrored to both disks, so deletion itself does not test redundancy. The disk-failure portion is what demonstrates RAID 1 availability.

Task 5 – RAID 5 vs RAID 10 for Cloud Storage

Question:

Summarize the main differences between RAID 5 and RAID 10 for a cloud storage service like Google Drive in 3 bullet points.

Answer:

- Capacity and redundancy: RAID 5 uses distributed parity and provides protection against one disk failure while using storage capacity efficiently. RAID 10 uses mirroring, so it provides strong redundancy but approximately half of the raw capacity is usable.
- Performance: RAID 10 generally provides better write performance because it does not require parity calculations. RAID 5 can provide good read performance but writes can have additional parity overhead.
- Failure and workload considerations: RAID 5 is useful when storage efficiency and protection from a single disk failure are important. RAID 10 is better suited to workloads requiring high performance and frequent writes, such as databases and virtualization. A large cloud service may choose different storage architectures depending on workload, cost, performance, and availability requirements.